

ECON 101

Introduction to Microeconomics

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These notes explain Chapters 1–3 of the lecture slides, preserving their examples and numerical data. The aim is to make the reasoning and its conditions explicit, rather than to introduce a separate mathematical model. Definitions and qualifications are the note-taker's exposition; the connecting lines in redrawn graphs are illustrative.

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§1 The Four Core Principles of Economics

An *economy* allocates scarce resources: what to produce, how to produce it, who does the work, and who receives the output. The slides use the stranded group in *Lord of the Flies* to illustrate these questions. Economics begins with individual decisions and studies how they fit together. The four principles below provide a way to reason about those decisions.

§1.1 The Cost–Benefit Principle

Definition 1.1.1 (Economic Surplus)

The *economic surplus* from a decision is its benefit minus its cost, measured on a common scale and relative to the same alternative.

$$\text{Economic surplus} = \text{benefit} - \text{cost}.$$

The cost–benefit principle says to choose an option over its alternative when the benefit is at least the cost. Equality means indifference; strictly greater benefit means a strict gain. With several mutually exclusive options, a positive surplus is not enough: another option may have a larger surplus.

Definition 1.1.2 (Willingness to Pay)

Willingness to pay is the greatest amount a person would pay to obtain a benefit, or to avoid a cost. It gives nonmonetary consequences, such as time saved, a dollar valuation.

A valuation is not necessarily a payment. An hour saved can be valuable even when no money changes hands. The slides also note that benefits can go to someone else and outcomes can be uncertain; assigning values in those situations requires more care than the simple examples here. Costs and benefits are *economic incentives* that influence choices. The principle is a guide to reasoning, not a claim that people always calculate or act this way.

Example 1.1.3 (Gains from Trade)

A buyer values a good at v , a seller's relevant cost is c , and they trade at price p . Relative to no trade, their surpluses are

$$\text{Buyer's surplus} = v - p, \quad \text{seller's surplus} = p - c.$$

Both weakly gain when $c \leq p \leq v$; both strictly gain when $c < p < v$.

Adding the surpluses cancels the payment: total surplus is $v - c$. Price determines how the gain is divided between these two parties. This explains why an exchange can benefit both, rather than being a zero-sum transfer. The comparison assumes no additional transaction costs and does not assess effects on third parties.

§1.2 The Opportunity Cost Principle

Definition 1.2.1 (Opportunity Cost)

The *opportunity cost* of a choice is the value of the next best feasible alternative forgone. Its *financial cost* is the money paid out of pocket; financial cost can miss valuable uses of time or resources.

Resources are *scarce*: using time, money or energy one way prevents using the same resource another way. Ask “this option, or what?” The relevant alternative is the next best one, not

the sum of every possible use of the resource. In accounting for costs, do not count the same forgone value twice.

For an hour of gaming, the major cost can be the next best use of the hour, rather than the hardware or software. An activity with no explicit payment can therefore have a substantial opportunity cost.

Definition 1.2.2 (Sunk Cost)

A *sunk cost* has already been incurred and cannot be recovered by changing the current decision. It does not distinguish the current options.

Ignoring it does not mean the money was unimportant. It means that choosing one option rather than another cannot change that cost. Subtracting the same amount from every option leaves their ranking unchanged.

Example 1.2.3 (Attending a Lecture)

The slides allocate \$4,500 tuition across five courses, each with 24 lectures: $4500/(5 \cdot 24) = \$37.50$ per lecture.

This is an average allocation, not the additional cost of attending. Once tuition is nonrefundable, skipping the lecture does not recover \$37.50. Compare the benefit of attending with the next best use of those 80 minutes, such as earning wages.

§1.3 The Production Possibilities Frontier

A production possibilities frontier shows the trade-offs imposed by limited resources. It describes what can be achieved, not which combination a person should prefer.

Definition 1.3.1 (Production Possibilities Frontier)

A combination of outputs is *feasible* if available resources and technology can produce it. It is *efficient* if one output cannot be increased without reducing another. The *production possibilities frontier* (PPF) consists of the efficient feasible combinations.

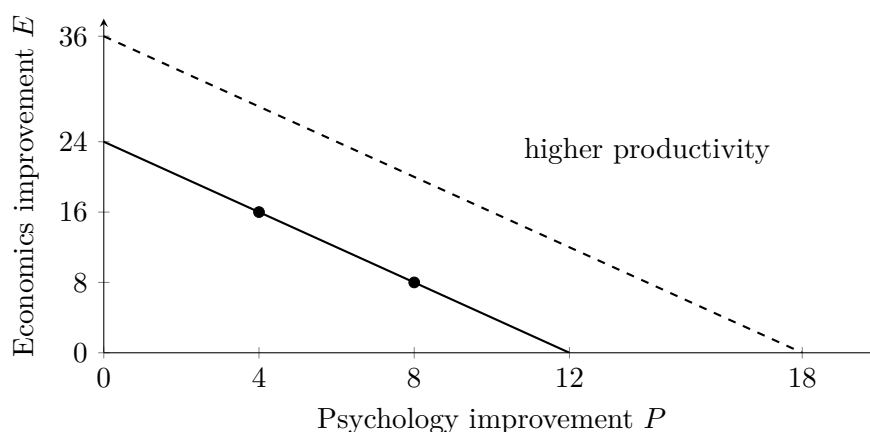
Points inside the frontier leave room to improve an output without sacrificing the other. Points outside it cannot be achieved with the current resources and technology.

Example 1.3.2 (Three Hours of Study)

The slides allow three hours per night for economics and psychology. Each economics hour adds eight grade points; each psychology hour adds four.

Economics hours	Psychology hours	Econ improvement	Psych improvement
3	0	24	0
2	1	16	4
1	2	8	8
0	3	0	12

These are grade improvements, not final grades. With psychology improvement P horizontally and economics improvement E vertically, the frontier is $E = 24 - 2P$ for $0 \leq P \leq 12$. Moving from $(P, E) = (4, 16)$ to $(8, 8)$ gains four psychology points and loses eight economics points. The opportunity cost of one psychology point is therefore two economics points.



The absolute slope measures the trade-off in vertical-axis output per unit of horizontal-axis output. Here it is constant because the assumed returns to each study hour are constant. A curved PPF can have a changing trade-off.

The dashed line reproduces the slides' productivity improvement. Increasing both hourly gains by 50% gives a frontier from (0, 36) to (18, 0). More combinations become feasible, although the opportunity cost remains two economics points per psychology point. An outward shift does not necessarily mean the trade-off is unchanged; it happens to be unchanged here.

§1.4 The Marginal Principle

Binary choices ask “which option?” Quantity choices ask “how much?” The marginal principle breaks the latter into comparisons of adjacent quantities.

Definition 1.4.1 (Marginal Benefit and Marginal Cost)

The *marginal benefit* of an additional unit is the increase in total benefit caused by that unit. Its *marginal cost* is the increase in total cost. For the k -th unit,

$$MB(k) = TB(k) - TB(k - 1), \quad MC(k) = TC(k) - TC(k - 1).$$

Buying or producing that unit changes surplus by $MB(k) - MC(k)$. This is a comparison of *changes*, not average benefit or average cost.

The rational rule. Continue while an additional unit's marginal benefit is at least its marginal cost. To make this a reliable stopping rule, marginal net benefit must not rise again after becoming negative. A common sufficient condition is that $MB - MC$ is nonincreasing across the feasible units. Then surplus rises with positive increments, stays unchanged with zero increments, and falls with negative increments.

An equality $MB = MC$ means the additional unit gives no extra surplus; including or excluding it can both be optimal. In a discrete problem, the sign may jump from positive to negative without ever reaching equality. The slides' “continue until they are equal” is therefore best understood as this marginal comparison, rather than a requirement to find exact equality.

Example 1.4.2 (Nerida's Restaurant)

The slides list worker counts from two to seven. A meal sells for \$25, food costs \$10 per meal, each worker costs \$300, rent costs \$500, and Nerida's time is valued at \$1,000, all for the same operating period.

For M meals and q workers, total benefit is $25M$ and total cost is $10M + 300q + 1500$. The resulting slide table is:

Workers	Meals	TB	MB	TC	MC	Surplus
2	160	4000	—	3700	—	300
3	210	5250	1250	4500	800	750
4	250	6250	1000	5200	700	1050
5	280	7000	750	5800	600	1200
6	300	7500	500	6300	500	1200
7	310	7750	250	6700	400	1050

The fourth worker adds $250 - 210 = 40$ meals: marginal benefit is $25 \cdot 40 = 1000$, while marginal cost is $300 + 10 \cdot 40 = 700$. The corresponding marginal surplus is \$300.

Both MB and MC decrease in this example. The marginal surpluses are \$450, \$300, \$150, zero, and \$-150. Five and six workers both maximise surplus at \$1,200; the sixth adds exactly zero. The comparison is limited to the listed staffing choices, since the slides supply no other output data. Rent and the value of Nerida's time cancel between these choices because they are the same in each. This alone does not make them sunk in a separate decision about opening or closing the restaurant.

§1.5 The Interdependence Principle

Your best choice depends on its setting. The slides distinguish four sources of interdependence:

- (i) **Your other choices.** Income, time, energy and attention are shared. Choosing economics study hours leaves fewer psychology study hours.
- (ii) **Other people's choices.** People, firms and governments compete for scarce resources, affecting the opportunities available to one another.
- (iii) **Other markets.** Conditions in labour, housing and credit markets can affect choices elsewhere.
- (iv) **Time.** Past choices and expectations of future opportunities influence current choices.

The next chapters first hold these other conditions fixed to study a response to price, then consider what happens when those conditions change.

In practice

- **Identify the alternative.** Compare costs and benefits on the same basis; count each forgone value once and exclude unrecoverable past costs.
- **Compare increments.** Subtract adjacent totals, then compare marginal benefit with marginal cost. Check that stopping does not miss later gains.
- **Read a PPF.** Check feasibility and efficiency, then calculate what must be sacrificed for the desired improvement.

Exercises

- (i) (Slides.) Discuss opportunity-cost explanations for teenagers dating but rarely marrying; fewer stay-at-home mothers than 60 years ago; more Canadian births in September–October than February–March; and terminally ill patients trying unproven drugs. These are proposed applications, not facts established by the opportunity-cost principle alone.

§2 Demand and Consumer Choice

Demand describes what buyers would choose at different possible prices. It connects willingness to pay with quantity decisions, then combines individual choices into a market relationship.

§2.1 Individual Demand

Definition 2.1.1 (Individual Demand Curve)

An *individual demand curve* shows the quantity a buyer is willing and able to buy at each price, holding other relevant conditions fixed (*ceteris paribus*). Quantity is measured over a specified period.

We can write this quantity as $q^D(p)$. Price is the input, but economics graphs put price on the vertical axis and quantity on the horizontal axis. A demand curve describes a range of possible choices; a quantity demanded is one choice at one price.

Example 2.1.2 (Darren's Gasoline Demand)

The slides survey Darren about his daily gasoline use:

Price (\$/litre)	2.40	2.20	2.00	1.80	1.60
Quantity (litres/day)	1	2	3	5	7

These are hypothetical-price responses; connecting them does not establish an exact formula at unobserved prices.

The law of demand. Holding other conditions fixed, a lower price weakly increases quantity demanded, and a higher price weakly decreases it. “Downward sloping” allows a buyer’s quantity to stay unchanged over some price ranges. This is the course’s usual demand relationship, not a statement that price is the only influence on buying.

§2.2 The Rational Rule for Buyers

A *price taker* treats the price as given rather than choosing it. For such a buyer, an additional unit’s marginal cost is its price. The rule is to buy another unit when its marginal benefit is at least that price, provided successive marginal benefits diminish.

Definition 2.2.1 (Diminishing Marginal Benefit)

Diminishing marginal benefit means that successive additional units provide no greater benefit than preceding units, with other conditions fixed. It does not mean that total benefit decreases.

If the extra benefit stays positive but gets smaller, total benefit is still rising, just in smaller increments. The assumption makes it sensible to stop once a unit costs more than it is worth: later units will not restore a positive marginal surplus.

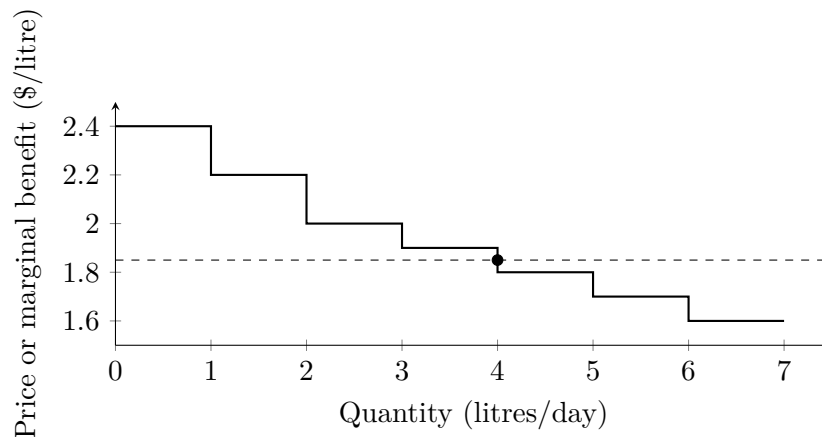
Example 2.2.2 (Darren's Ranked Uses)

The slides assign these benefits to successive daily litres:

Unit	Use	Marginal benefit (\$)
1	Drive to work instead of taking the bus	2.40
2	Drive to school instead of biking	2.20
3	Shop at No Frills instead of a dearer shop	2.00
4	Visit friends	1.90
5	Drive to the gym instead of jogging there	1.80
6	Drive for errands instead of walking	1.70
7	Take a scenic drive	1.60

At \$1.85 per litre, the first four uses are worth at least their cost. The fifth is worth only \$1.80, so it would reduce surplus by \$0.05; subsequent uses are worth still less. Darren therefore buys four litres in this example.

This explains the law of demand in the slide model: lowering the price makes more of the ranked uses worthwhile, while raising it rules some out. At \$1.80, the fifth litre adds exactly zero surplus. Including it reproduces the survey's five litres, but buying four would give the same surplus. We include units at equality when describing the tabulated choices.



The steps illustrate the slide's marginal values; four complete units are worthwhile at the dashed price line.

The slides say to buy until marginal benefit equals price. With discrete units, the precise stopping rule is that the last purchased unit is worth at least the price and the next is worth less. Exact equality need not occur. For smoothly divisible quantities, equality describes an interior optimum when marginal benefit falls through the price; a budget or quantity limit can instead make an endpoint optimal.

§2.3 Market Demand**Definition 2.3.1 (Market Demand)**

Market demand adds all individual quantities demanded at the same price. For N buyers,

$$Q^D(p) = \sum_{i=1}^N q_i^D(p).$$

This is called a *horizontal sum*, because quantity is on the horizontal axis.

If every buyer weakly reduces their quantity when price rises, adding those quantities gives the same direction for the market total. A price change can affect both existing customers' quantities and whether some potential customers buy at all. A customer moving from zero to positive purchases is included in this comparison.

Example 2.3.2 (Surveying 400 Consumers)

The slides scale a 400-person sample to a population of 40 million:

Price	Darren	Brooklyn	Sample total	Projected millions
1.60	7	4	1200	120
1.80	5	3	1100	110
2.00	3	2	1000	100
2.20	2	1	900	90
2.40	1	0	800	80

Prices are dollars per litre; quantities are litres per day, with the last column expressed in millions.

The scale factor is $40,000,000/400 = 100,000$. For example, $1000 \cdot 100,000 = 100$ million litres at \$2.00. The projection assumes that the sample represents the population; it is an estimate, not an exact population count.

§2.4 Movements Along Versus Shifts of Demand

Definition 2.4.1 (Movement and Shift)

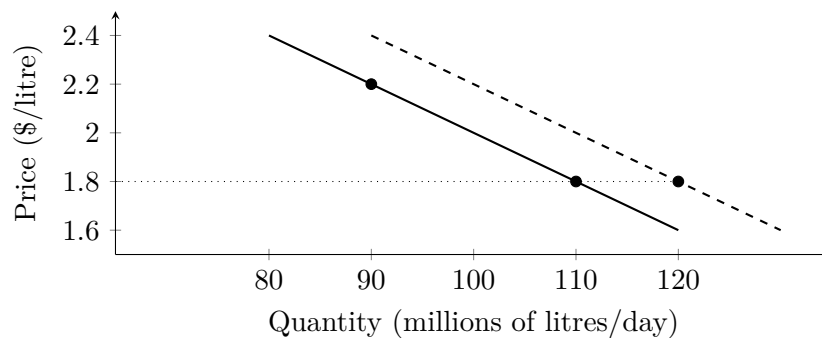
A change in the good's own price, holding other conditions fixed, is a *movement along the demand curve*; it changes *quantity demanded*. A change in another relevant condition can *shift the demand curve*: there is a different quantity demanded at the same price. This changes *demand*, meaning the relationship as a whole.

An increase in demand is a right shift: more is demanded at each price in the range being compared. A decrease is a left shift. The direction is read horizontally, even when the curve is downward sloping.

Example 2.4.2 (Two Comparisons)

On the original market curve, lowering price from \$2.20 to \$1.80 raises quantity from 90 to 110 million litres per day. The slide's increased-demand curve instead has 120 million litres at the same price of \$1.80, rather than 110 million.

The first changes price on one curve; the second compares curves at a fixed price. Thus “price fell, so demand increased” should be “price fell, so quantity demanded increased”, with other conditions fixed.



The solid curve connects the observations. The dashed curve illustrates increased demand, with the slide's comparison at \$1.80.

§2.5 The Six Demand Shifters

Definition 2.5.1 (Normal and Inferior Goods)

A *normal good* has higher demand when income increases; an *inferior good* has lower demand, holding price and other conditions fixed over the income range being considered.

“Inferior” describes the income response, not the good’s quality. Higher income does not automatically increase demand for every good.

Definition 2.5.2 (Complements and Substitutes)

Goods are *complements* when lowering the price of one raises demand for the other. They are *substitutes* when lowering the price of one lowers demand for the other, holding other relevant conditions fixed.

The six shifters in the slides are:

- (i) **Income.** The direction depends on whether the good is normal or inferior. A tax cut raising disposable income shifts gasoline demand right if gasoline is normal in the relevant range.
- (ii) **Preferences.** For example, a marketing campaign can make a brand more desirable and raise willingness to pay.
- (iii) **Prices of related goods.** A cheaper complement raises demand; a cheaper substitute lowers it. The price changing is the *other* good’s price, so this shifts the original good’s curve.
- (iv) **Expectations.** Beliefs about future prices can change purchases today, such as buying earlier in anticipation of a price increase.
- (v) **Congestion and network effects.** A *network effect* makes a good more valuable as more other people use it; a *congestion effect* makes it less valuable. These can change a buyer’s willingness to pay.
- (vi) **Type and number of buyers.** Adding buyers increases the market sum, holding existing buyers’ curves fixed. This direct aggregation effect changes market demand without changing those individual curves.

Congestion or network effects may additionally change individual curves; the last factor isolates the direct effect of adding buyers.

In practice

- **Buy another unit?** Compare its marginal benefit with price; check how a tie is treated.
- **Aggregate at one price.** Add quantities, with matching units and periods, rather than adding prices.
- **Movement or shift?** Ask whose price changed. The good’s own price gives a movement; other conditions can shift its curve.

§3 Supply and Producer Choice

Supply describes sellers' quantity decisions at different prices. The marginal principle still applies, but the additional benefit is now revenue from selling, while the additional cost comes from production.

§3.1 Individual Supply

Definition 3.1.1 (Individual Supply Curve)

An *individual supply curve* shows the quantity a seller is willing and able to sell at each price, holding other conditions fixed. Quantity is measured over a specified period; price is on the vertical axis.

Example 3.1.2 (Shell's Gasoline Supply)

The first supply graph in the slides gives:

Price (\$/litre)	1.60	1.80	2.00	2.20	2.40
Quantity (millions of litres/day)	8	9	10	11	12

Below \$1.60, the slides state that Shell supplies zero.

These data describe the example, but do not explain its jump from zero to eight million litres. They also do not establish quantities above \$2.40.

The law of supply. With other conditions fixed, a higher price weakly increases quantity supplied and a lower price weakly decreases it. “Upward sloping” allows unchanged quantities over some price ranges.

§3.2 Perfect Competition and Price Taking

Definition 3.2.1 (Perfect Competition)

The slide model of *perfect competition* assumes:

- (i) many buyers and sellers, each small relative to the market, with no barriers to entry or exit;
- (ii) an identical product across firms;
- (iii) perfect information;
- (iv) zero costs for buyers to switch sellers.

A *price taker* treats the market price as given when choosing output.

With identical products and costless switching, charging more than other sellers loses buyers. A firm that can sell its chosen output at the going price has no reason to charge less for that output. The slides also appeal to free entry and competition to explain why profitable undercutting opportunities would not persist.

We use price taking as the model's assumption, rather than treating this informal argument as a complete proof about every market. Perfectly competitive markets are uncommon in practice; the simplification lets us study output choice without also modelling a firm's influence on price.

§3.3 The Rational Rule for Sellers

For a price taker, the marginal benefit of selling one more unit is the market price. Compare it with the additional cost of producing that unit.

Definition 3.3.1 (Fixed and Variable Costs)

A *fixed cost* does not vary with the output level under consideration; a *variable cost* does.

Total cost can be written as

$$TC(q) = F + VC(q),$$

where F is fixed cost and $VC(q)$ is variable cost.

When the same F is incurred at both quantities, it cancels in their comparison. Thus the marginal cost of the k -th unit is

$$MC(k) = VC(k) - VC(k - 1).$$

The relevant cost includes opportunity costs, not only explicit payments. *Profit*, measured as revenue minus these economic costs, is the producer's economic surplus.

The seller's rule. Sell another unit when its price is at least its marginal cost. If marginal cost is nondecreasing, units cease to be worthwhile once their marginal cost exceeds price; later units will not reverse that comparison. This explains the law of supply in this model: raising price makes more units worthwhile, while lowering it makes fewer worthwhile.

For discrete units, the last produced unit can have marginal cost below price while the next has marginal cost above it. A unit with marginal cost equal to price adds zero profit. The slide's equality rule describes a smooth interior choice, rather than an equality every discrete answer must satisfy.

Remark (Fixed Does Not Mean Sunk). Fixed and variable describe dependence on output. Sunk and avoidable describe whether a decision can change the payment. A fee paid only if the firm operates can affect whether it should produce at all, even if it does not vary with positive output. First find the best output if operating, then compare its profit with not operating. Ignore a fixed payment only when it is common to the options actually being compared.

§3.4 Diminishing Marginal Product

Definition 3.4.1 (Marginal Product)

A *production function* relates inputs to output. The *marginal product* of labour is the additional output from an additional unit of labour, with other inputs fixed. *Diminishing marginal product* means successive extra units of labour add smaller amounts of output.

For example, adding workers while keeping kitchen equipment fixed can make each additional worker less productive. Total output may still rise; it is the increment that shrinks.

If each extra worker costs the same wage but adds less output, the labour cost per additional output unit rises. The slides identify this as one reason marginal cost tends to increase. Rising input prices provide another reason. The argument keeps other conditions fixed; diminishing labour product by itself need not describe a firm's costs if it can change all inputs or improve technology.

§3.5 The Production Example

Example 3.5.1 (Labour with Fixed Capital)

The slides use $Y = \sqrt{LK}$, where L is labour, K is capital and Y is output. With capital fixed at $K = 4$, this becomes $Y = 2\sqrt{L}$.

Labour L	0	1	2	3	4	5	6
Output Y	0	2	2.83	3.46	4.00	4.47	4.90
Marginal product	—	2	0.83	0.64	0.54	0.47	0.43

The third labour unit adds $2\sqrt{3} - 2\sqrt{2} \approx 0.64$ output units. Subtracting rounded entries gives $3.46 - 2.83 = 0.63$, which differs because of rounding. Calculate the change from exact values before rounding. The table shows total output increasing while each additional labour unit adds less.

To calculate the cost of a specified output, solve the same relation for labour: $L = Y^2/4$. At a wage of \$15 per labour unit, the labour cost is $15L = 3.75Y^2$. The slides' second table is:

Output Y	0	1	2	3	4	5	6
Labour L	0	0.25	1	2.25	4	6.25	9
Cost (\$)	0	3.75	15	33.75	60	93.75	135
Marginal cost (\$)	—	3.75	11.25	18.75	26.25	33.75	41.25

Capital remains 4, and its cost is omitted from this table. Adding a common fixed capital cost would raise the cost totals but leave the marginal-cost row unchanged.

The third output unit requires $2.25 - 1 = 1.25$ more labour units, costing $15 \cdot 1.25 = \$18.75$. It costs more than the second output unit, which needs only $1 - 0.25 = 0.75$ more labour units and costs \$11.25. This is the link between diminishing marginal product and rising marginal cost.

The two tables change different quantities: the first adds one labour unit at a time, while the second adds one output unit at a time and permits fractional labour. Their marginal rows therefore measure different things. For a seller at price \$15 per output unit, the second unit is worth producing because its marginal cost is \$11.25; the third is not because it costs \$18.75. Within this table, two units maximise profit even though no marginal cost equals price.

§3.6 Market Supply

Definition 3.6.1 (Market Supply)

Market supply adds all sellers' quantities at the same price. For M sellers, $Q^S(p) = \sum_{j=1}^M q_j^S(p)$. For identical sellers, $Q^S(p) = Mq^S(p)$.

This is a horizontal sum. If each seller supplies weakly more at a higher price, so does the sum. A higher price can also make previously inactive potential sellers supply a positive quantity.

Example 3.6.2 (Ten Identical Sellers)

The aggregation slide gives:

Price (\$/litre)	1.60	1.80	2.00	2.20	2.40
Individual quantity	6	8	10	12	14
Market quantity	60	80	100	120	140

Quantities are millions of litres per day. For example, at \$2.20, market quantity is $10 \cdot 12 = 120$ million litres.

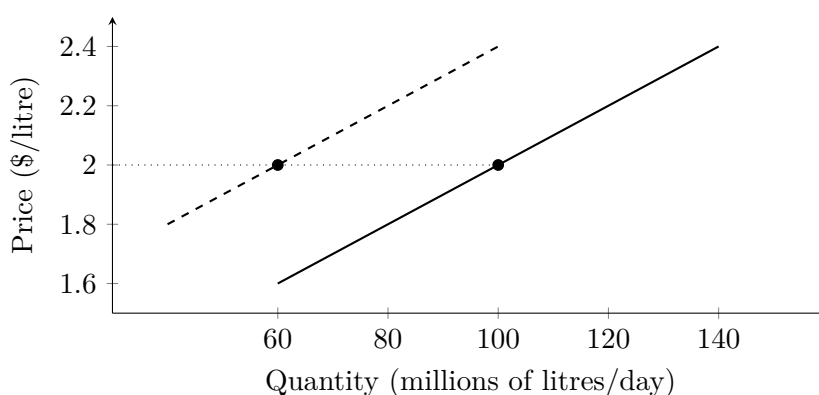
This slide uses different individual quantities from Shell's first graph in Example 3.1.2. Treat it as a separate numerical illustration.

§3.7 Movements Along Versus Shifts of Supply

Definition 3.7.1 (Movement and Shift)

Changing a good's own price with other conditions fixed gives a *movement along the supply curve*, changing *quantity supplied*. Changing another relevant condition can *shift the supply curve*, changing *supply*: a different quantity is supplied at the same price. An increase in supply shifts right; a decrease shifts left.

On the original market curve, raising gasoline price from \$1.80 to \$2.20 raises quantity from 80 to 120 million litres per day. In the crude-oil example, gasoline price stays at \$2.00, but a higher input cost lowers supply from 100 to 60 million litres per day. The first is a movement; the second is a left shift.



The dashed curve illustrates the slide's decrease in supply. The essential comparison is at the fixed gasoline price of \$2.00.

§3.8 The Five Supply Shifters

- (i) **Input prices.** Higher input prices raise marginal cost and reduce the units worth supplying at a given output price. Crude oil is an input to gasoline, so its price rise shifts gasoline supply left.
- (ii) **Productivity and technology.** Improvements that lower the cost of additional output shift supply right. Examples include new equipment, better management or production methods, and learning by doing.
- (iii) **Prices of related outputs.** *Substitutes in production* are alternative outputs made with the same inputs; a higher price for one encourages reallocating inputs to it, reducing supply of the other. *Complements in production* are produced together; a higher price for one can encourage joint production and increase supply of the other. These are production relationships, distinct from consumer substitutes and complements.
- (iv) **Expectations.** If output can be stored, an expected future price increase can encourage withholding it today. This is more relevant for nonperishable gasoline than fresh fish; storage costs also matter.
- (v) **Type and number of sellers.** Adding sellers increases the market sum, holding existing individual curves fixed. A merger or Competition Bureau ruling can change the seller population and other market conditions, potentially affecting supply.

The distinction between movement and shift depends on which price changes. Gasoline's own price is on its supply graph; the price of crude oil is a condition behind that graph.

§3.9 Buyers and Sellers Compared

Both use the marginal principle. Buyers compare the value of another unit with its price; sellers compare its price with its additional production cost.

	Buyer	Price-taking seller
Marginal benefit	Value of extra unit	Price received
Marginal cost	Price paid	Cost of extra production
Take another unit if	$MB \geq p$	$p \geq MC$
Usual quantity response	Falls as price rises	Rises as price rises
Underlying explanation	Diminishing marginal benefit	Rising marginal cost
Market curve	Sum of buyers' quantities	Sum of sellers' quantities

These marginal stopping rules rely on the stated pattern of benefits or costs. The equality formulations are useful for smooth interior choices, but should not replace the comparisons in the slides' discrete tables.

In practice

- **Specify the decision.** Choosing output with costs already committed differs from deciding whether to operate. Cancel only common costs.
- **Keep the increment straight.** An extra worker and an extra unit of output are different changes; identify which the table measures.
- **Identify the price.** Own-price changes give movements along a curve; input prices and other conditions can shift it.

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